Technology Innovation Hub for IoT and IoE (TIH-IoT), IIT Bombay
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Pest Detection Model

A 13-class ResNet-50 classifier for multi-crop pest identification, deployed in the Mahavistaar application

Project: Multi-Crop Disease and Pest Detection using Deep Learning and Vision-Language Models
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Abstract. This report describes the pest detection model developed under the TIH-IoT CHANAKYA Fellowship. A ResNet-50 network, trained with two-phase transfer learning on the PoCRA pest dataset, classifies a photograph into one of 13 classes: twelve pest categories and healthy plants. On a held-out test set of 4,538 images the model reaches 98.3% accuracy and a weighted precision, recall and F1-score of 0.983, with every class at or above 0.96 F1. Grad-CAM++ heat maps are used to inspect the image regions that drive each prediction. The model is deployed in the Mahavistaar farmer-advisory application and was demonstrated at the AI Summit under Hon. Chief Minister Devendra Fadnavis.

1 Introduction

Manual scouting for crop pests is slow and subjective, and by the time symptoms are obvious the yield loss is often already locked in. Image-based identification allows a pest to be recognised from a photograph taken in the field. The objective of this work was to build a pest detection model covering 13 categories across multiple crops using the PoCRA dataset, and to integrate it into a deployable advisory application. The work was carried out between 1 December 2025 and 30 September 2026 and the objective was met in full.

The model distinguishes the following classes: Aphid, Cotton Leaf Roller, Dusky Cotton Bug, Fall Army Worm, Hairy Caterpillar, Jassid, Mealybug, Pink Bollworm, Pod Borer, Red Cotton Bug, Thrips, White Fly, and Healthy.

2 Dataset

The PoCRA pest dataset contains 30,226 labelled images in 13 classes. It was divided once into training, validation and test sets in a 70/15/15 ratio (Table 1). The validation set was used to monitor training and to select the best checkpoint; the test set was held out until the final evaluation.

Table 1. Dataset split.

Split	Images	Share
Training	21,168	70.0%
Validation	4,520	15.0%
Test	4,538	15.0%
Total	**30,226**	**100%**

Class sizes are uneven. In the test set they range from 78 images (Cotton Leaf Roller) to 921 images (Jassid). To prevent the larger classes from dominating training, the loss function uses inverse-frequency class weights computed from the training set, so that rarer classes receive larger weights, from 0.38 for Jassid to 4.47 for Cotton Leaf Roller (listed in Table 3).

3 Method

3.1 Network architecture

The classifier is a ResNet-50 initialised with ImageNet weights. Its original 1,000-way output layer is replaced by a new head consisting of dropout (rate 0.3) and one linear layer from 2,048 features to 13 classes. ResNet-50 is built from bottleneck residual blocks whose skip connections keep gradients stable through 50 layers, which makes it a robust backbone for transfer learning. The complete network has 23,534,669 parameters.

3.2 Two-phase transfer learning

Training proceeds in two phases (Figure 1). In Phase 1 the backbone is frozen and only the new head is trained; this involves 26,637 parameters, 0.11% of the network, for up to 50 epochs with early stopping. Warming up the head first prevents the randomly initialised layer from sending large gradients into the pretrained features. In Phase 2 the last two residual stages, Layer-3 and Layer-4, are unfrozen (about 22.1 million parameters, 94% of the network) so that high-level features adapt to pest imagery. The stem, Layer-1 and Layer-2 remain frozen because the generic edge and texture features they compute transfer directly.



Figure 1. Trainable parameters in each phase of training, shown per stage of ResNet-50.

3.3 Training configuration

The loss is cross-entropy with the class weights described above. Training stops when validation loss has not improved by at least 0.0001 for five consecutive epochs, and the checkpoint with the best validation accuracy is kept together with the final model and the full training history. Table 2 lists the settings.

Table 2. Training configuration.

Setting	Value
Backbone	ResNet-50, ImageNet-pretrained
Head	Dropout 0.3, linear layer 2,048 to 13
Loss	Cross-entropy, inverse-frequency class weights
Optimiser	Adam, learning rate 0.001, weight decay 0.0001
Batch size	32 (662 iterations per epoch)
Epochs	Up to 50 per phase
Early stopping	Patience 5, minimum improvement 0.0001
Hardware	NVIDIA GPU (CUDA), 4 data-loading workers

4 Results

The model was evaluated once on the 4,538 test images. It reaches 98.3% accuracy and weighted precision, recall and F1-score of 0.983, corresponding to about 4,460 correct predictions. Mealybug is classified perfectly (F1 1.000) and five classes exceed 0.99 F1. All 13 classes reach at least 0.96 F1, including the smallest class, Cotton Leaf Roller (78 images, F1 0.994). The mean of the 13 per-class F1-scores (macro-F1) is approximately 0.985. Per-class results are shown in Figure 2 and Table 3.

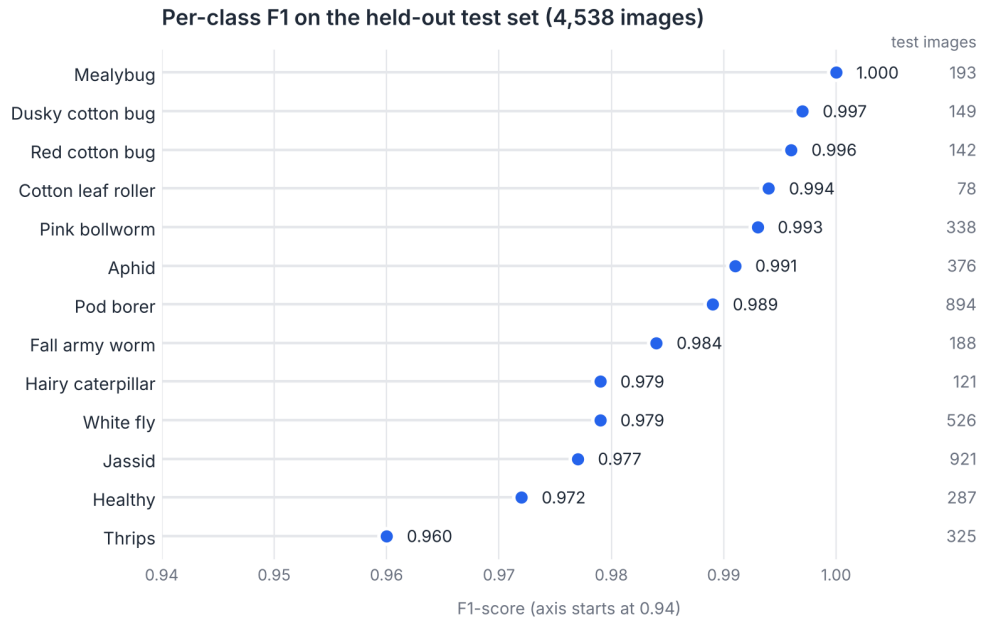


Figure 2. F1-score per class on the held-out test set. The number of test images per class is shown on the right.

Table 3. Per-class results on the test set (support is the number of test images) and the class weight used in the training loss.

Class	Precision	Recall	F1-score	Support	Class weight
Aphid	0.987	0.995	0.991	376	0.9127
Cotton Leaf Roller	1.000	0.987	0.994	78	4.4734
Dusky Cotton Bug	0.993	1.000	0.997	149	2.3262
Fall Army Worm	0.984	0.984	0.984	188	1.8214
Hairy Caterpillar	0.983	0.975	0.979	121	2.8973
Healthy	0.969	0.976	0.972	287	1.2115
Jassid	0.978	0.975	0.977	921	0.3776
Mealybug	1.000	1.000	1.000	193	1.8275
Pink Bollworm	0.991	0.994	0.993	338	1.0425
Pod Borer	0.991	0.988	0.989	894	0.3917
Red Cotton Bug	1.000	0.993	0.996	142	2.5089
Thrips	0.954	0.966	0.960	325	1.0855
White Fly	0.981	0.977	0.979	526	0.6649
Weighted average	**0.983**	**0.983**	**0.983**	**4,538**	

5 Explainability

To show which part of a photograph drives a prediction, the system produces Grad-CAM++ heat maps. The method takes the gradient of the predicted class score with respect to the feature maps of the last residual block, weights each feature map, keeps the positive evidence and overlays the result on the photograph; red regions contribute most to the decision. In the example in Figure 3 the strongest activation lies on the small white insect on the leaf surface, so the prediction rests on the pest and not on the background.

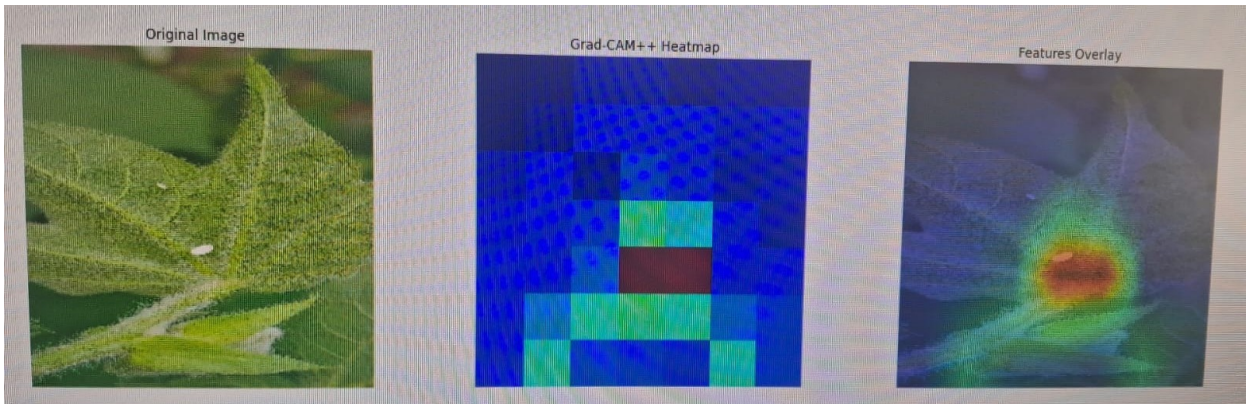


Figure 3. Original photograph, Grad-CAM++ heat map and overlay for a cotton leaf with an insect pest.

A Gradio web application lets a user upload a photograph and view the heat map together with the class probabilities. A confidence threshold of 0.7 is used so that uncertain predictions can be reported as “Unknown” instead of a wrong pest name.

6 Deployment

The trained model is served behind a prediction API (Figure 4) and is live inside the Mahavistaar farmer-advisory application. It was showcased at the AI Summit under Hon. Chief Minister of Maharashtra Devendra Fadnavis. Each request is validated, preprocessed, checked for the presence of a crop, classified, filtered by confidence and returned as a structured response. The code is written in PyTorch and torchvision, with scikit-learn for evaluation and OpenCV for the heat-map overlays.

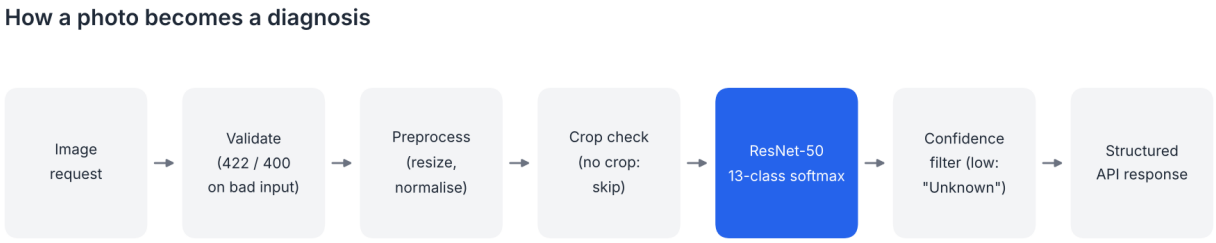


Figure 4. Prediction service flow, from request validation to the structured response.

7 Conclusion

A ResNet-50 classifier trained with two-phase transfer learning identifies 13 pest and healthy classes with 98.3% accuracy on unseen images, explains its decisions with Grad-CAM++, and is in use in the Mahavistaar application. Planned next steps are to extend the pest taxonomy as further labelled images become available, to feed field images back into training, and to benchmark lighter backbones (ResNet-18, MobileNetV2, EfficientNet-B0, all supported by the same code base) for on-device inference.